Research Statement

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My research lies at the intersection of agricultural economics, environmental economics, and transportation economics. I use quasi-experimental methods to evaluate how public policies shape household welfare, environmental outcomes, and resource allocation, with a focus on Taiwan and broader developing-economy contexts. My work spans three interconnected themes.

1. Agricultural Energy Transition and Climate Policy

Why it matters. The agricultural sector faces a dual imperative: adapting to climate change while contributing to decarbonization. Governments have increasingly used both demand-side instruments (e.g., rooftop solar incentives for farms) and supply-side border measures (e.g., carbon pricing in trade) to achieve climate goals, yet the welfare consequences for agricultural producers and downstream markets remain poorly understood. Rigorous empirical evidence on these effects is essential to avoid policies that inadvertently undermine food security or international competitiveness.

What my research does. I study the farm-level economics of clean energy adoption using Taiwan's comprehensive agriculture census data. In Liou, Chang, and Begun (2025a), I find that rooftop solar installation increases farm revenue on average, suggesting that energy transition subsidies can generate genuine co-benefits for crop producers. A complementary study of poultry producers (Lee, Liou, and Chang 2025b) confirms that solar adoption is a win–win strategy for chicken farms—improving both profitability and environmental performance—while also revealing heterogeneous effects across farm size. Shifting from domestic to international dimensions, Liou, Chang, and Begun (2025b) develops a theoretical framework to examine how the European Union's Carbon Border Adjustment Mechanism (CBAM) reshapes global trade flows and competitive dynamics, identifying conditions under which the mechanism improves or worsens welfare in exporting nations. Together, these studies map out a coherent research programme on how energy and climate policies propagate through agricultural supply chains.

2. Transportation Externalities, Road Safety, and Urban Mobility

Why it matters. Urban transportation generates large externalities—traffic fatalities, greenhouse gas emissions, and congestion—that impose enormous social costs. Demand for evidence on targeted policy levers is high: transit subsidies, traffic regulations, and land-use rules each have the potential to reshape travel behavior at scale, but causal identification has historically been difficult because policies are rarely assigned randomly. The COVID-19 pandemic further disrupted commuting patterns in ways that are not yet fully understood, creating a rare natural experiment on behavioral inertia in transport demand.

What my research does. I exploit quasi-random variation in policy timing and geographic rollout to identify causal effects. Liou et al. (2025) shows that the introduction of monthly transit passes in Taiwan's major cities significantly increases public-transport ridership and reduces carbon emissions, consistent with a downward-sloping demand curve for travel that responds to price even among regular commuters. Liou and Chang (2026a) evaluates the removal of hook-turn restrictions at busy intersections and finds a meaningful reduction in serious traffic accidents, providing rare causal evidence on how intersection-level regulation affects road safety outcomes. On the demand side, Chang et al. (2021) documents a sharp and persistent decline in Taipei Metro ridership during the COVID-19 pandemic, pointing to the role of risk perception—beyond formal lockdown policies—in reshaping daily mobility. Extending this line of work to event-driven transport demand, Liou and Chang (2026b) quantifies the elevated accident risk associated with large-scale religious tourism, offering actionable estimates for traffic management planning around recurrent mass-gathering events.

3. Environmental Externalities, Natural Resources, and Consumer Behavior

Why it matters. Economic activity generates diffuse environmental externalities—from air pollution to biodiversity loss—whose effects on households are often invisible in aggregate statistics. Understanding how households and farms respond to these externalities, and to macroeconomic shocks like pandemics, is essential for designing interventions that improve welfare without distorting resource allocation. This is particularly salient in mixed-use rural landscapes, where agricultural livelihoods interact closely with natural ecosystems and cultural practices that generate both positive and negative spillovers.

What my research does. Lee, Liou, and Chang (2025a) examines how forest biodiversity shapes the distribution of farm revenue in adjacent agricultural areas of Taiwan, finding that higher forest diversity is associated with a more equal distribution of farm earnings—suggesting that ecosystem services act as a natural equalizer among smallholder farms. On the pollution side, Chang, Liou, and Meyerhoefer (2024) identifies a significant spike in air pollution around Taiwan's Tomb-Sweeping Day, attributing the effect to the ritual burning of paper offerings; the study links religious-cultural externalities to quantifiable public health costs and offers a template for assessing pollution from periodic cultural events elsewhere in East Asia. Expanding the focus to macroeconomic disruptions and service sector responses, Liou, Kim, and Chang (2026) investigates hotel recovery strategies in the post-pandemic era, providing empirical evidence on how revenue, employment, and labor market rigidity interact as firms adapt to large-scale structural shocks. Finally, Liou, Chang, and Just (2024) applies a Bayesian change-point model to high-frequency credit card transaction data to trace how Taiwanese consumers adjusted spending across categories in response to the COVID-19 outbreak, revealing asymmetric patterns of precautionary saving and substitution that standard survey methods would have missed.

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